

```
from google.colab import files
uploaded = files.upload()
```

No file chosen

Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

```
import pandas as pd

df = pd.read_csv("1 iris.csv")
df.head()
```

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column          Non-Null Count  Dtype
---  -
 0   sepal_length    150 non-null   float64
 1   sepal_width     150 non-null   float64
 2   petal_length    150 non-null   float64
 3   petal_width     150 non-null   float64
 4   species         150 non-null   object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
```

```
df.isnull().sum()
```

```
0
sepal_length    0
sepal_width     0
petal_length    0
petal_width     0
species         0

dtype: int64
```

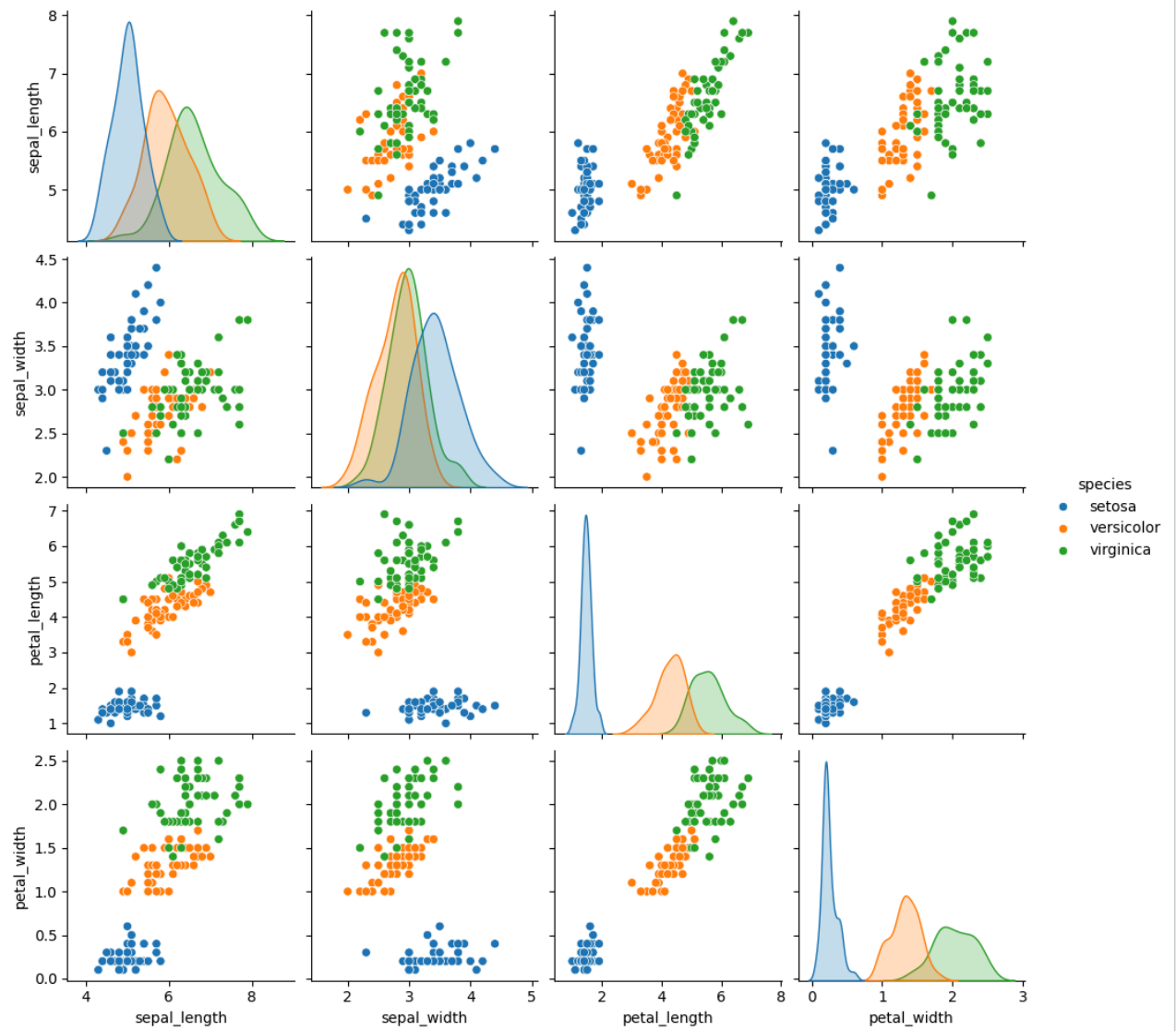
```
df.describe()
```

	sepal_length	sepal_width	petal_length	petal_width
count	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.054000	3.758667	1.198667
std	0.828066	0.433594	1.764420	0.763161
min	4.300000	2.000000	1.000000	0.100000
25%	5.100000	2.800000	1.600000	0.300000
50%	5.800000	3.000000	4.350000	1.300000
75%	6.400000	3.300000	5.100000	1.800000
max	7.900000	4.400000	6.900000	2.500000

```
df.drop_duplicates(inplace=True)
```

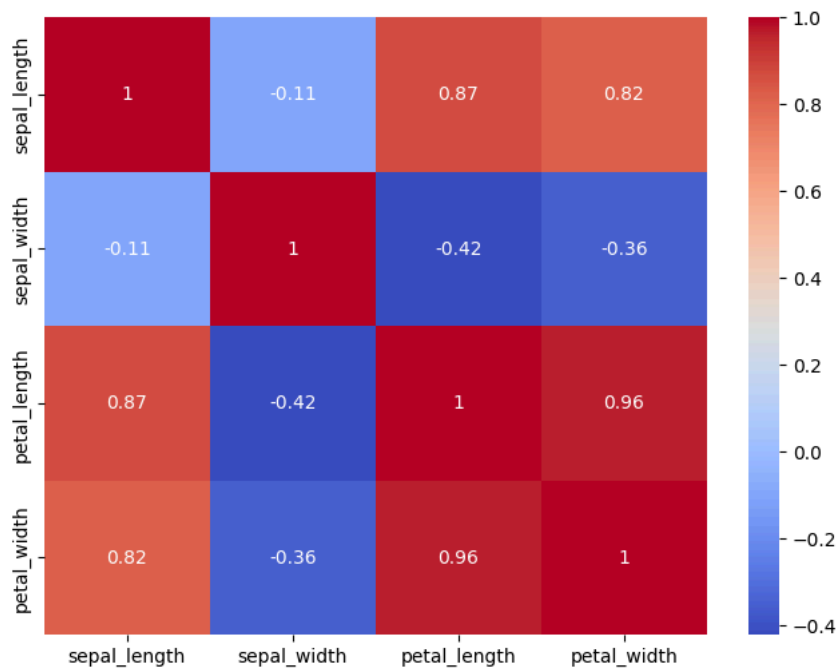
```
import seaborn as sns
import matplotlib.pyplot as plt

sns.pairplot(df, hue='species')
plt.show()
```

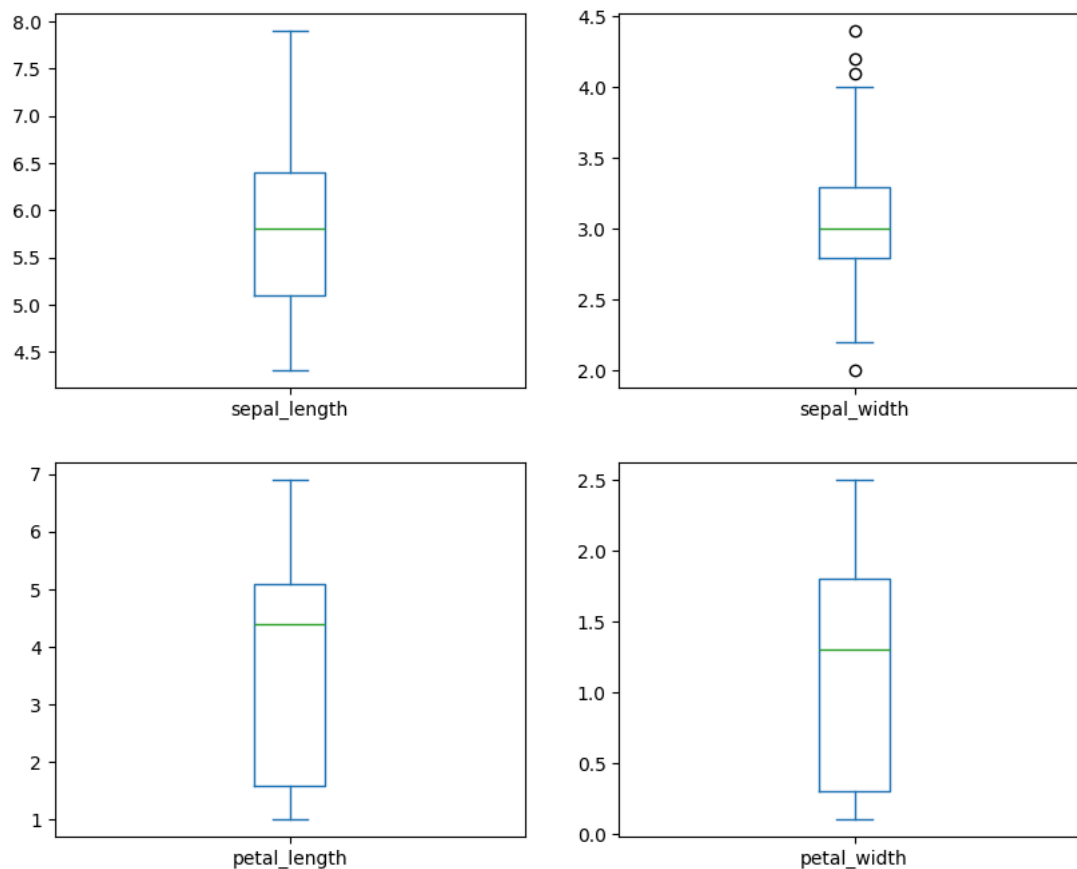


```
numeric_df = df.select_dtypes(include=['float64', 'int64'])
```

```
plt.figure(figsize=(8,6))
sns.heatmap(numeric_df.corr(), annot=True, cmap='coolwarm')
plt.show()
```



```
df.plot(kind='box', subplots=True, layout=(2,2), figsize=(10,8))
plt.show()
```



```
df.to_csv("cleaned_iris.csv", index=False)
```

```
from google.colab import files
files.download("cleaned_iris.csv")
```

```
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score
```

```
X = df.drop("species", axis=1)
y = df["species"]
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

model = RandomForestClassifier()
model.fit(X_train, y_train)

predictions = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, predictions))
```

Accuracy: 0.9333333333333333

```
from google.colab import files
files.upload()
```

[Show hidden output](#)

```
import pandas as pd

df = pd.read_csv("4) house Prediction Data Set.csv")
df.head()
```

	0.00632	18.00	2.310	0	0.5380	6.5750	65.20	4.0900	1	296.0	15.30	396.90	4.98	24.00
0	0.02731	0.00	7.070	0	0.4690	6.4210	78...							
1	0.02729	0.00	7.070	0	0.4690	7.1850	61...							
2	0.03237	0.00	2.180	0	0.4580	6.9980	45...							
3	0.06905	0.00	2.180	0	0.4580	7.1470	54...							
4	0.02985	0.00	2.180	0	0.4580	6.4300	58...							

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
df.info()
df.head()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 505 entries, 0 to 504
Data columns (total 1 columns):
#   Column
---  ---
0    0.00632  18.00    2.310  0    0.5380  6.5750  65.20  4.0900  1  296.0  15.30  396.90  4.98  24.00
dtypes: object(1)
memory usage: 4.1+ KB
```

	0.00632	18.00	2.310	0	0.5380	6.5750	65.20	4.0900	1	296.0	15.30	396.90	4.98	24.00
0	0.02731	0.00	7.070	0	0.4690	6.4210	78...							
1	0.02729	0.00	7.070	0	0.4690	7.1850	61...							
2	0.03237	0.00	2.180	0	0.4580	6.9980	45...							
3	0.06905	0.00	2.180	0	0.4580	7.1470	54...							
4	0.02985	0.00	2.180	0	0.4580	6.4300	58...							

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
df = pd.read_csv("4) house Prediction Data Set.csv", delim_whitespace=True)
df.head()
```

```
/tmp/ipykernel_3212/2029134270.py:1: FutureWarning: The 'delim_whitespace' keyword in pd.read_csv is deprecated and will be
df = pd.read_csv("4) house Prediction Data Set.csv", delim_whitespace=True)
```

	0.00632	18.00	2.310	0	0.5380	6.5750	65.20	4.0900	1	296.0	15.30	396.90	4.98	24.00
0	0.02731	0.0	7.07	0	0.469	6.421	78.9	4.9671	2	242.0	17.8	396.90	9.14	21.6
1	0.02729	0.0	7.07	0	0.469	7.185	61.1	4.9671	2	242.0	17.8	392.83	4.03	34.7
2	0.03237	0.0	2.18	0	0.458	6.998	45.8	6.0622	3	222.0	18.7	394.63	2.94	33.4
3	0.06905	0.0	2.18	0	0.458	7.147	54.2	6.0622	3	222.0	18.7	396.90	5.33	36.2
4	0.02985	0.0	2.18	0	0.458	6.430	58.7	6.0622	3	222.0	18.7	394.12	5.21	28.7

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
df.columns = [
    "CRIM", "ZN", "INDUS", "CHAS", "NOX", "RM", "AGE",
    "DIS", "RAD", "TAX", "PTRATIO", "B", "LSTAT", "PRICE"
]
```

```
df.info()
df.head()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 505 entries, 0 to 504
```

```
Data columns (total 14 columns):
```

#	Column	Non-Null Count	Dtype
0	CRIM	505 non-null	float64
1	ZN	505 non-null	float64
2	INDUS	505 non-null	float64
3	CHAS	505 non-null	int64
4	NOX	505 non-null	float64
5	RM	505 non-null	float64
6	AGE	505 non-null	float64
7	DIS	505 non-null	float64
8	RAD	505 non-null	int64
9	TAX	505 non-null	float64
10	PTRATIO	505 non-null	float64
11	B	505 non-null	float64
12	LSTAT	505 non-null	float64
13	PRICE	505 non-null	float64

```
dtypes: float64(12), int64(2)
```

```
memory usage: 55.4 KB
```

	CRIM	ZN	INDUS	CHAS	NOX	RM	AGE	DIS	RAD	TAX	PTRATIO	B	LSTAT	PRICE
0	0.02731	0.0	7.07	0	0.469	6.421	78.9	4.9671	2	242.0	17.8	396.90	9.14	21.6
1	0.02729	0.0	7.07	0	0.469	7.185	61.1	4.9671	2	242.0	17.8	392.83	4.03	34.7
2	0.03237	0.0	2.18	0	0.458	6.998	45.8	6.0622	3	222.0	18.7	394.63	2.94	33.4
3	0.06905	0.0	2.18	0	0.458	7.147	54.2	6.0622	3	222.0	18.7	396.90	5.33	36.2
4	0.02985	0.0	2.18	0	0.458	6.430	58.7	6.0622	3	222.0	18.7	394.12	5.21	28.7

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
from sklearn.model_selection import train_test_split
```

```
X = df.drop("PRICE", axis=1)
y = df["PRICE"]
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

```
from sklearn.model_selection import train_test_split
```

```
X = df.drop("PRICE", axis=1)
y = df["PRICE"]
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

```
from sklearn.linear_model import LinearRegression
```

```
model = LinearRegression()
model.fit(X_train, y_train)
```

▼ LinearRegression ⓘ ?

```
LinearRegression()
```

```
pred = model.predict(X_test)
```

```
from sklearn.metrics import mean_squared_error, r2_score
```

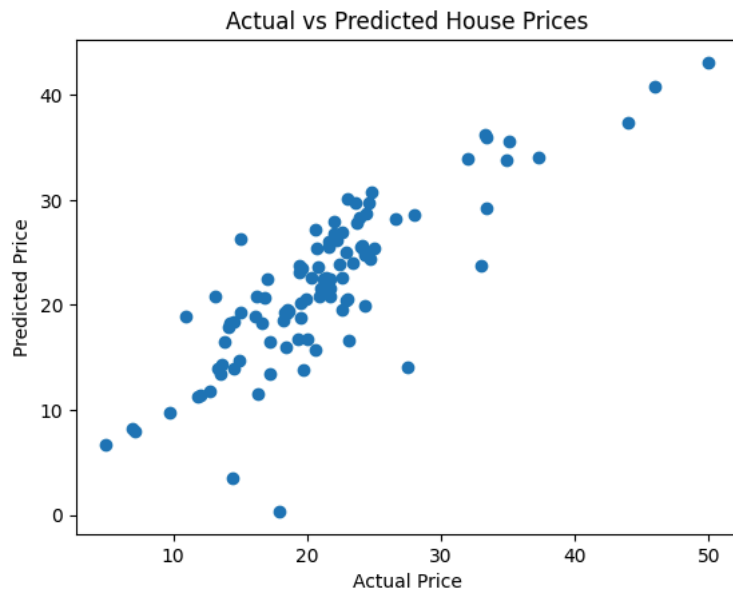
```
print("MSE:", mean_squared_error(y_test, pred))
print("R2 Score:", r2_score(y_test, pred))
```

```
MSE: 19.379041116137028
R2 Score: 0.652220188866615
```

```
import matplotlib.pyplot as plt
```

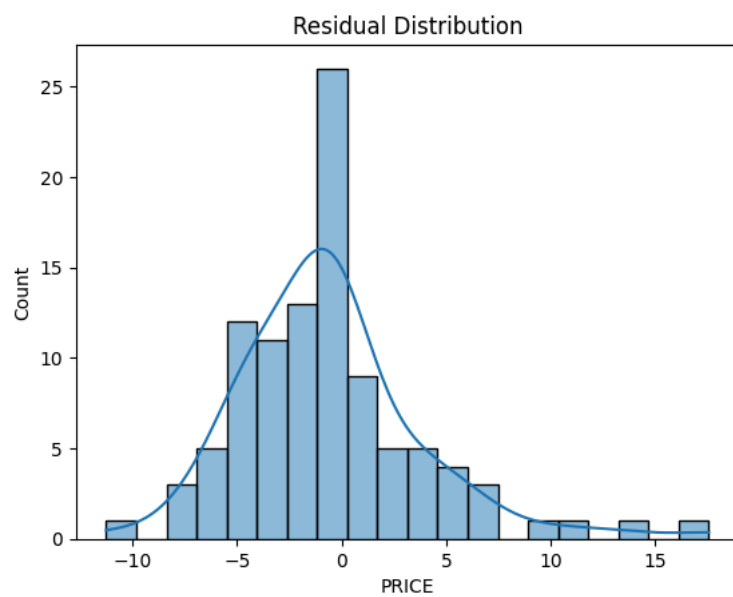
```
plt.scatter(y_test, pred)
```

```
plt.xlabel("Actual Price")
plt.ylabel("Predicted Price")
plt.title("Actual vs Predicted House Prices")
plt.show()
```



```
import seaborn as sns

sns.histplot(y_test - pred, bins=20, kde=True)
plt.title("Residual Distribution")
plt.show()
```



```
df_result = pd.DataFrame({
    "Actual": y_test,
    "Predicted": pred
})

df_result.to_csv("house_predictions.csv", index=False)
```

```
from google.colab import files
files.download("house_predictions.csv")
```